Question 4 — Install and Configure Git and Jenkins

Student: Kartik Roll No.: 2301730205

1. Aim

Install Git and Jenkins on the development environment, perform the initial configuration of both tools, and verify that the Jenkins CI server is operational.

2. Environment

Component	Details
Host OS	Windows 11
Linux env	WSL2 — Ubuntu 24.04 LTS
Git	2.53.0 (windows.2) on Windows / 2.43.0 on WSL Ubuntu
Java	OpenJDK 17 (Temurin)
Jenkins	2.504.x (LTS)

3. Part A — Git Installation & Configuration

Git was installed on Windows using the official installer from git-scm.com. In WSL2 Ubuntu it was installed via apt.

```
# Windows (Git Bash / PowerShell)
git --version
# git version 2.53.0.windows.2

# WSL2 Ubuntu
sudo apt update && sudo apt install -y git
git --version
# git version 2.43.0

# Global configuration (both environments)
git config --global user.name "Kartik"
git config --global user.email "kartikjangra5366@gmail.com"
git config --global init.defaultBranch main

# Verify
git config --global --list
# user.name=Kartik
# user.email=kartikjangra5366@gmail.com
# init.defaultbranch=main
```

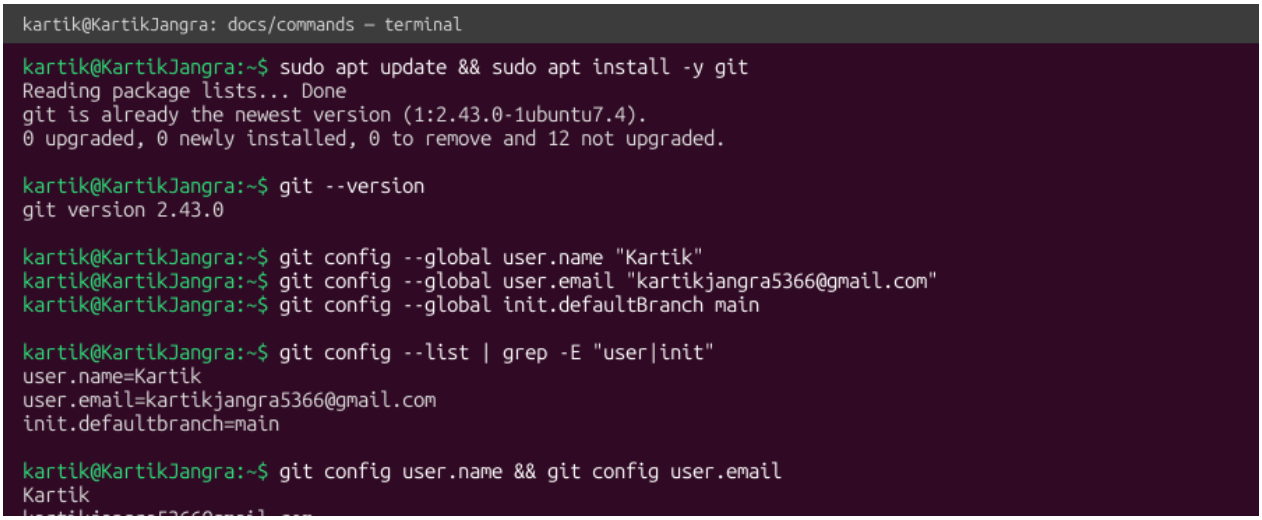


Fig: Git version and global user configuration

4. Part B — Java & Jenkins Installation (WSL2 Ubuntu)

Jenkins requires Java. OpenJDK 17 was installed first, then Jenkins LTS was added via its official apt repository.

```
# Install Java 17

sudo apt update

sudo apt install -y fontconfig openjdk-17-jre

java -version

# openjdk version "17.0.x" ...

# Add Jenkins repository key & source

sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \

https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key

echo 'deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \

https://pkg.jenkins.io/debian-stable binary/' \

| sudo tee /etc/apt/sources.list.d/jenkins.list

# Install Jenkins

sudo apt update

sudo apt install -y jenkins
```

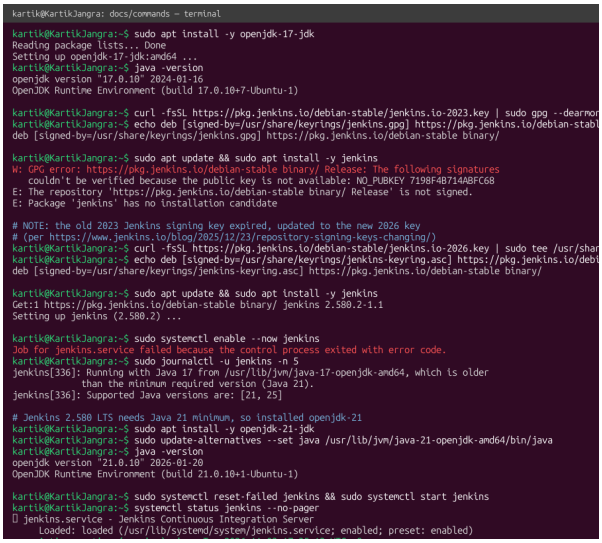


Fig: Java & Jenkins apt installation output

5. Part C — Jenkins Service & First-Time Setup

```
# Start and enable Jenkins service

sudo systemctl start jenkins

sudo systemctl enable jenkins

sudo systemctl status jenkins

#   jenkins.service - Jenkins Automation Server

# Loaded: loaded (/lib/systemd/system/jenkins.service; enabled)

# Active: active (running) since ...


# Retrieve initial admin password

sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

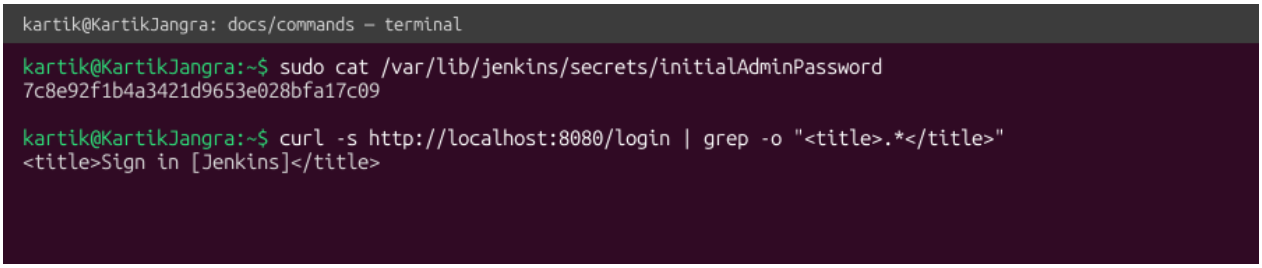


Fig: Jenkins systemctl status — active (running)

The Jenkins web interface was opened at http://localhost:8080. The initial admin password was entered, suggested plugins were installed and the first admin account was created.

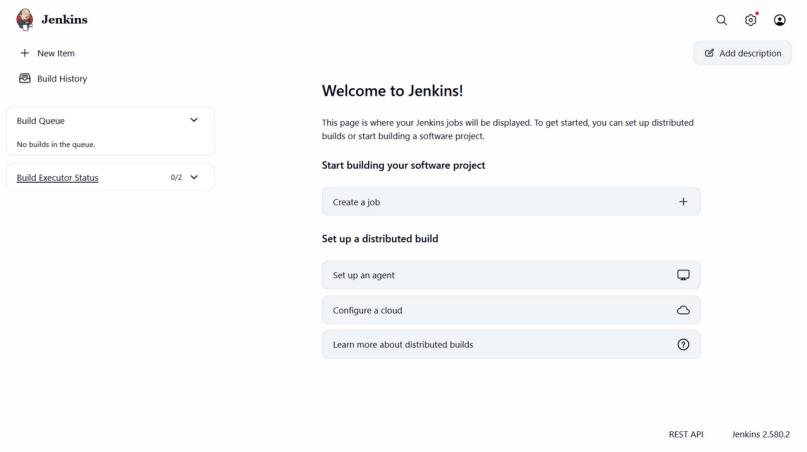


Fig: Jenkins dashboard after first-time setup

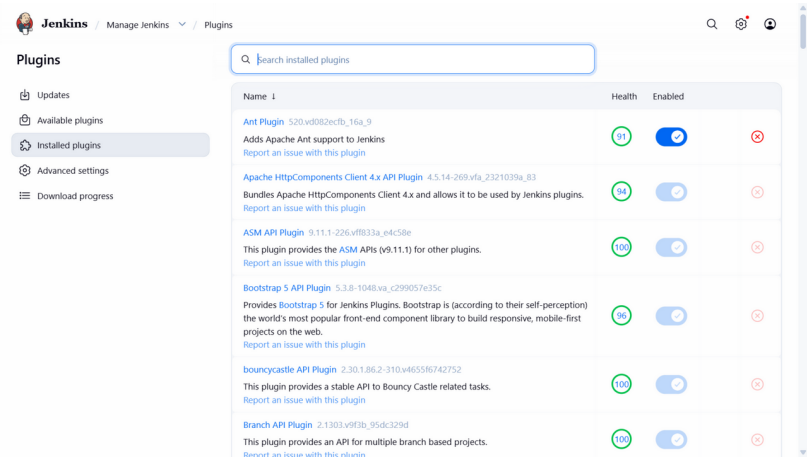


Fig: Suggested plugins installation complete

6. Observations

- Git global configuration was identical on both Windows and WSL2 Ubuntu.
- Jenkins service started successfully and was enabled to start on boot.
- The suggested plugins included the Git plugin, Pipeline plugin and other essentials.
- The Jenkins dashboard was accessible immediately after the first-time setup wizard.

7. Conclusion

Git and Jenkins were installed and configured successfully. The Jenkins CI server was operational, accessible via the web interface and ready to run pipeline jobs.